DraftFM: Zero-Shot Drafting of Unseen *Magic: The Gathering* Sets from Public Card Features

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Abstract

Draft assistants for *Magic: The Gathering* require weeks of post-release human pick data before they can model a new set; on release day they have nothing. We train DraftFM, a 2.2M-parameter pick model that represents every card as a frozen vector of public information (structured Scryfall features plus a sentence embedding of rules text) and carries no set-identity parameters, so it can score a set the moment its card list is public. Trained on 149.4M human picks from 28 sets of 17Lands public data, the model reaches 54.3% expert-pick agreement zero-shot on three held-out sets (48.9% / 57.5% / 56.5% on BRO / TMT / SOS), 78.7% of the accuracy of per-set models trained directly on each target set, and above every published approach that is feasible on release day. The headline evaluation is pre-registered and frozen before the test set exists: one inference pass over the first public snapshot of MSH, a licensed-IP set whose pick data nothing in the pipeline has touched ([PENDING: F-full MSH top-1]% expert top-1). We release the code, the frozen protocol, model weights, and per-pick prediction files.

54.3%	**78.7%**	**2.2M params**
zero-shot expert-pick agreement on three unseen sets (§6)	of a per-set-trained ceiling's accuracy, with zero picks from the target set	trains from scratch in 1.8 h on one workstation (§4.3)

1 Introduction

Drafting in *Magic: The Gathering* is a sequential game of imperfect information: eight players pass packs of cards, each privately assembling a deck one pick at a time. Strong pick models exist. Supervised models trained on human draft logs reach 48.7–71% agreement with expert picks within a set [5, 6, 10], but they are set-specific. Each model encodes cards as vocabulary indices, learns from picks made in that set, and transfers nowhere. Because public draft logs appear on a delay, a data-driven assistant for a new set cannot exist until roughly two weeks after release [5]. On release day, the state of the art is a spreadsheet of hand-tuned expert ratings [7].

This paper asks how far pure transfer can go. DraftFM is a single pick model trained across every public 17Lands draft set, in which a card is nothing but a frozen vector of public information:

structured features parsed from its Scryfall record and a sentence embedding of its rules text. The model has no set-identity parameters and no per-card weights. Whatever it knows about a set it must infer from the cards in the pack and the drafter's accumulating pool, not from a memorized identity. That inference transfers to a set released tomorrow; a memorized card vocabulary cannot.

The evaluation is designed against a failure mode endemic to this setting: a "zero-shot" number quietly tuned on its own test set. Our protocol (§5) was frozen, in a tagged public commit, before the test set existed. The test set is MSH, a licensed-IP set whose public draft data had not been published at the freeze; no training, tuning, or metric selection touches MSH pick data, and the headline number is produced by a single pre-registered inference pass over the first public snapshot. All model selection happens on three development sets (BRO, TMT, SOS) held out of training, chosen so our numbers are directly comparable to the only published zero-shot benchmark [3] (same BRO holdout) and to our own strongest per-set ceilings.

Zero-shot on the development trio, DraftFM agrees with expert picks 54.3% of the time (48.9% / 57.5% / 56.5% on BRO / TMT / SOS): 78.7% of what per-set models trained on each target achieve, and above every published number attainable on release day (expert-tuned ratings 44.5% [10], an hour-zero rarity heuristic 42.8%) as well as zero-shot GPT-4o (43% [2]). On MSH the frozen evaluation gives [PENDING: F-full MSH top-1]% expert top-1 agreement.

Contributions:

- **A cross-set model and recipe.** A 2.2M-parameter two-tower architecture over frozen card features that trains on 149.4M picks in about 1.8 hours on a single consumer workstation (§4).
- **A frozen zero-shot benchmark.** A pre-registered, tamper-evident protocol for day-one draft evaluation, plus the first multi-set zero-shot numbers that use no post-release information of any kind (§5, §6).
- **A scaling study.** Zero-shot accuracy as a function of training-set count from 1 to 28 sets, with seed and set-composition variance (§6).
- **A licensed-IP shift analysis.** Pre-registered ablations designed to isolate what text embeddings contribute when a set's names and flavor come from a licensed universe rather than in-universe *Magic: The Gathering* (§7).
- **Release.** Code, the frozen protocol, data manifests, model weights, and per-pick prediction files for every number in this paper (Appendix B).

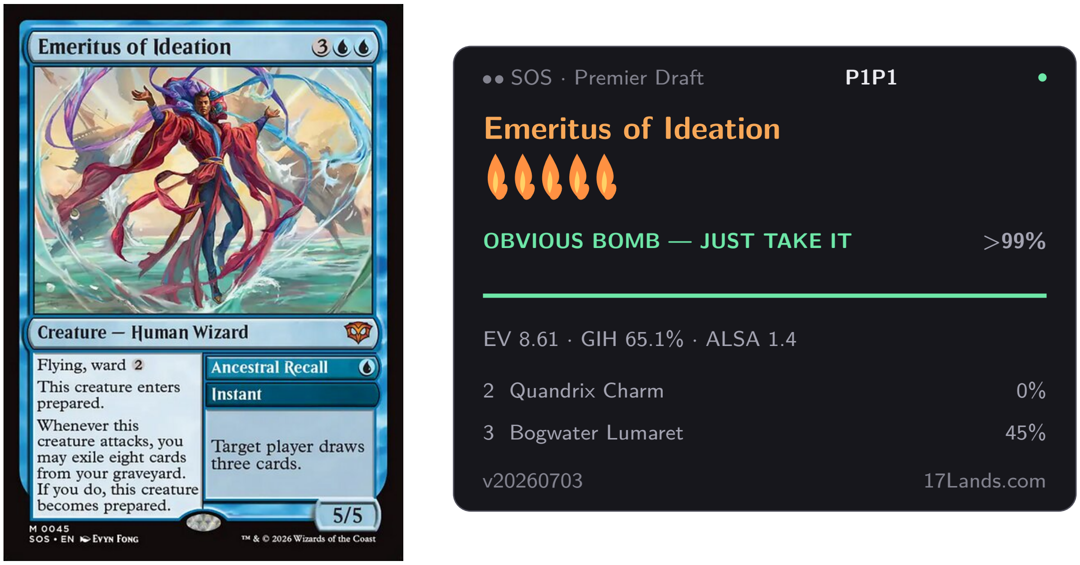

Figure 1: The pick-scoring interface, on an actual first pick. The card (left) is Scryfall's unmodified scan, copyright and artist line intact; the panel (right) mirrors the deployed overlay's live "verdict" view (§4.2) and every number in it is real, computed by the shipped per-set scoring model (Table 1's per-set ceiling) over an actual thirteen-card SOS pack, empty pool. SOS is a public development set, never the frozen MSH test set (§5). This figure illustrates the interface, not a headline result. Numbers regenerated by `figures/make_verdict_panels.py`.

2 Related work

Generalised card representations. Bertram, Fürnkranz, and Müller [3] established the zero-shot drafting problem, and their paper remains the only published quantitative result on it. They replace one-hot card identities with concatenated feature vectors: hand-engineered attributes plus a Sentence-BERT embedding [8] of card text (1306-d), a learned autoencoding of card images (1024-d), and 16 post-release usage statistics from 17Lands. This lives inside a contextual-preference-ranking model trained with a triplet loss. Two of their findings shape our design: feature representations match one-hot accuracy within a set (nothing is lost by generalising), and multi-set pretraining, not the representation alone, is what buys transfer. Pretrained on 13 sets and evaluated on held-out BRO, their best representation reaches 55.44% pick agreement. That representation includes the usage-statistics block, which is computed from human play after a set's release; the number is therefore an anchor for zero-shot *card generalisation* rather than for day-one deployment, where such statistics cannot exist. The same paper's day-one-feasible representation (features only) reaches 33.6–35.6% on unseen sets under single-set training. No multi-set features-only number is published. Filling that gap (multi-set training, public card information only) is this paper's headline experiment, and we adopt their held-out BRO as a development set so the comparison is on the same holdout. A companion paper [4] improves the within-set triplet objective with an adapted InfoNCE loss but reports no unseen-set results.

Within-set pick models. Ward et al. [10] compare heuristic and neural drafters on M19 and report the two most useful reference points below the data-driven regime: an expert-tuned rating bot at 44.5% top-1 agreement and a random agent at 22%. Their neural drafter (48.7%) and its successors, the statistical-drafting MLPs deployed at statisticaldrafting.com [5] (∼70%) and Czerner's puder transformer [6] (71%), define the within-set state of the art, and the statistical-drafting architecture is the direct lineage of the per-set ceiling models we compare against. All are one-hot models over a fixed card vocabulary; none can score an unseen set. The deployment gap is explicit: statistical-drafting documents that no model can ship until 17Lands data appears, roughly two weeks post-release [5], and Draftsim's day-one ratings are written by human experts [7].

LLMs as drafters. UrzaGPT [2] evaluates large language models on draft picks. Zero-shot GPT-4o reaches 43%, narrowly above the hour-zero rarity heuristic we measure (§6). We still mark it non-day-one in Table 2: NEO shipped years before GPT-4o's training cutoff, so its pretraining corpus cannot be shown free of the set's own post-release strategy discussion, a leak a frozen feature-only encoder cannot have. Providing full rules text in the prompt lowers GPT-4o's accuracy further. Open 7–8B models fail to pick legal cards until LoRA-tuned on a million picks from the target set, after which they still trail small supervised MLPs at orders of magnitude more inference cost. We take the implied guidance: use a frozen text *embedding* as one feature block, not a language model as the policy.

Text-grounded transfer in card games. Cardsformer [11] encodes Hearthstone card descriptions with a language model so a gameplay policy generalises to unseen cards. It is the closest cross-game precedent for text-mediated transfer, applied to gameplay rather than drafting.

3 Data

Corpus. Training data is the complete 17Lands public draft-data corpus [1], spanning every set the model has to generalise across: 31 expansions from STX (April 2021) through SOS (June 2026), Premier (best-of-one) and Traditional (best-of-three) draft where published, 59 (set, format) files, roughly 160M pick rows. Each row is one human pick: the pack contents, the drafter's pool, the card taken, and two coarse skill covariates (recent game-count bucket and win-rate bucket). Files were fetched once from the public S3 bucket, per the 17Lands usage guidelines (bulk downloads only), and frozen by sha256 and ETag in a data manifest (content hash `443bd25f72ee`); the manifest is released with the paper.

Two public files are structurally excluded from the corpus, recorded in the corpus registry: OM1 (a pick-two format whose rows carry two picks and whose pool columns undercount them, and whose digital-only cards largely do not exist on Scryfall) and the Powered Cube (a curated 545-card singleton pool with no expansion structure). A third, the VOW QuickDraft file (human picks inside bot pods, off-distribution pack dynamics), is held out of the default corpus for a distributional reason rather than a structural one: it is one of the two pre-registered flexibility-clause extras (§5) that may join F-full's training data if a dev-trio ablation shows a gain. Everything else is kept, including digital-only and remaster sets.

Normalization. The corpus spans three schema eras and several quirks, all normalised at ETL time: the 2021 files are gzipped tarballs rather than plain gzipped CSVs, and the oldest era reports match-based rather than game-based skill buckets (harmonised; identical in best-of-one). Five sets are missing the first pick of each draft in the source logs (an annotation, not an exclusion), and packs are 13, 14, or 15 picks depending on the set (a model input, §4). Card names are joined to Scryfall by normalised name with an explicit alias table; unresolved names fail the build rather than featurizing as zeros. Per set we retain every valid pick and tag it with the drafter's skill covariates: training filters no one out. Skill is a conditioning input ("decision + tag by venue").

Splits. Within each (set, format) shard, drafts are split by a hash of the draft identifier (`crc32 % 1000 < 50`, $\sim$5% validation). Early stopping uses only this within-training-set validation split. The *expert slice* used for all headline evaluation is picks by drafters with win-rate bucket ≥ 0.55 and game-count bucket ≥ 100, the same filter the per-set ceiling models train and evaluate on, so ceiling comparisons are population-matched.

The held-out sets. Four sets never contribute a pick to model selection. BRO, TMT, and SOS are the development trio (§5): all model selection keys on zero-shot accuracy over these three, computed from *F-dev*, which trains on none of them. Once selection is frozen, the headline model (*F-full*) retrains on all 31 sets, dev trio included: only MSH stays unseen by every model in the battery. MSH, the test set, is stricter still: a registry-level gate (`EVAL_ONLY`) makes every training and feature-fitting code path refuse MSH pick data outright, and at the protocol freeze its public data did not yet exist. MSH *card* features from Scryfall are permitted. That is the zero-shot contract: public card information only, no human behavioral data of any kind.

4 Method

4.1 Cards as frozen feature vectors

Every card is a fixed 775-dimensional vector computed from public information alone, frozen before training.

Structured features (391-d). Parsed from the card's Scryfall record: mana value (scaled + bucketed one-hot, 10), mana-cost pips including hybrid/Phyrexian/X structure (10), colors (8), color identity (5), supertypes (3), card types (9), creature subtypes (top-128 vocabulary + an unmatched count, 129), power / toughness / loyalty each as (scaled, missing, star) triples (9), rarity (5), keyword abilities (166 slots + unmatched count, 167), layout including back-face flags for double-faced cards (10), and text-shape features: length, line count, and 24 regex flags over rules text for common functional categories (removal, card draw, counterspells, tokens, ramp, ...) (26). Double-faced and split cards take numeric blocks from the front face. Subtype and keyword vocabularies are counted over *training* sets only; a registry gate refuses to fit vocabularies on the held-out set. The resulting featurizer manifest is content-hashed (`e1313cadd03b`) and released; held-out cards featurize through the frozen manifest, with out-of-vocabulary subtypes and keywords absorbed by the unmatched counters. Scalars are clipped to $[0, 1]$; parse failures set a missing indicator, never a silent zero.

Text embedding (384-d). The card's type line and rules text, embedded with a frozen `bge-small-en-v1.5` sentence encoder [12] (L_2-normalised). The card's own name (and, for legendary cards, its short name) is masked to a placeholder token before embedding, so the encoder cannot key on names it may know from pretraining, a deliberate handicap that matters for licensed-IP sets (§7). Parenthetical reminder text is kept: for a brand-new mechanic it is the only definition the model will ever see. Back faces are appended for double-faced cards.

4.2 Architecture

DraftFM is a two-tower scorer with 2.2M parameters (Figure 2). Its two design invariants operationalize the zero-shot contract (§3):

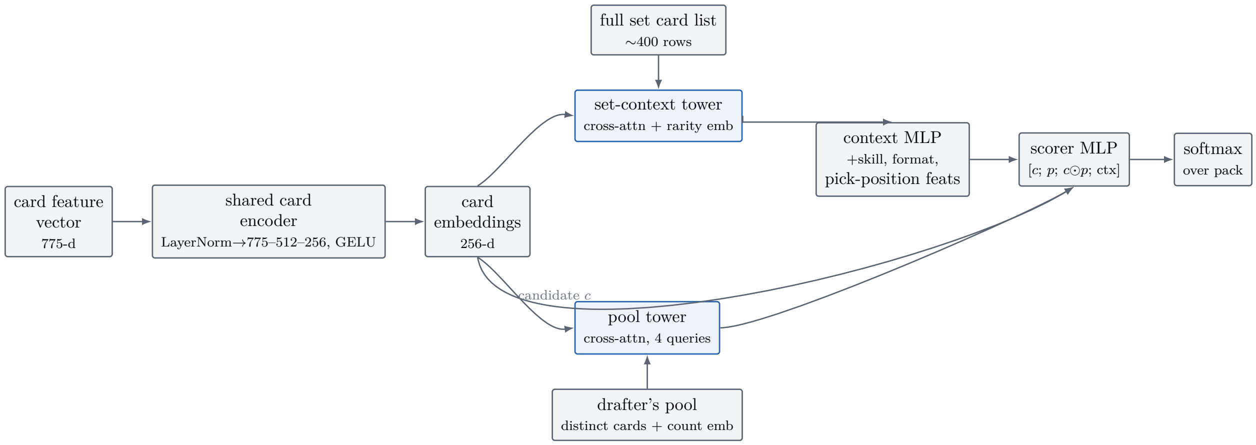

Figure 2: The two-tower scorer (§4.2). A shared card encoder embeds every card once per (set, format) step; a pool tower summarises the drafter's held cards, a set-context tower summarises the full card list, and a per-candidate scorer combines the candidate embedding c, the pool summary p, their elementwise product, and the context vector into a softmax over the pack.

1. **No set identity.** There is no set embedding and no per-card parameter anywhere. The only way the model can know what set it is drafting is a *set-context tower* that reads the set's full card list, plus the drafter's accumulating pool. "Blue is the aggressive color in this set" must be inferred from the card list, not memorized as a set identifier. Whether the tower itself turns out to be necessary for that inference, or whether the invariant holds just as well without it, is addressed later in this section.

2. **Skill conditions the scorer, not the cards.** Card semantics are skill-invariant; the pick policy is not. Skill-bucket embeddings enter only the context pathway. Serving conditions on a fixed high-skill bucket; training sees every pick tagged with its drafter's bucket.

A shared *card encoder* (LayerNorm → 775–512–256 MLP, GELU) maps feature vectors to d=256 embeddings. Because training batches are homogeneous in (set, format), the encoder runs once per step over the set's full card list (∼400 rows) and everything downstream gathers rows from that table. The *pool tower* represents the drafter's pool as the set of distinct cards held, each embedding translated by a learned count embedding (counts capped at 8), pooled by cross-attention from four learned queries (4 heads); an empty pool attends over a learned null token. The *set-context tower* applies the same query-pooling to the entire card list (with a rarity embedding added) to produce the set summary. A context MLP combines the set summary with skill and format embeddings, seven pick-position features (pack/pick indices, pool size), and four set-shape scalars (card-list size, picks-per-pack indicator) into a 64-d context. Each candidate c in the pack is scored by an MLP over $[c;\ p;\ c \odot p;\ \text{ctx}]$, where p is the pool summary and $c \odot p$ the elementwise interaction; a softmax over the pack's real slots gives the pick distribution.

The set-context tower's contribution is tested directly: the A-noctx ablation (Table 4, §7) removes it and scores higher on all three dev sets, and that tower-free configuration – not the one just described – is what F-full and the frozen MSH evaluation run. The first invariant holds either way, since no set-identity parameter exists in either configuration; dropping the tower only means the shipped model's sole route to set-relative signal is the pool tower, each candidate's own features, and the four set-shape scalars above, never a read of the rest of the card list.

4.3 Training

One recipe for every run in the paper (fixed before the scaling and ablation studies, since the protocol forbids per-rung tuning). Batches of 8,192 picks are drawn from one (set, format) shard at a time, shards sampled proportionally to $n_{\text{picks}}^{0.5}$, an interpolation between pick-uniform (which lets the largest sets dominate) and set-uniform (which oversamples tiny sets). AdamW (β_1=0.9, β_2=0.98, weight decay 0.01), peak learning rate 10^{-3}, 2,000-step linear warmup, cosine decay; cross-entropy with label smoothing 0.05 renormalised over real pack slots. The presentation budget is four epochs. Within-training validation (never the dev sets) is evaluated every 2,000 steps and training stops after three non-improving evaluations. Seed 17 throughout. Seed variance is measured at two scaling rungs (§5).

Training runs on a single Apple M3 Ultra workstation (96 GB unified memory) in PyTorch 2.12 eager mode on the MPS backend, at ∼42,750 picks/s. The full 149.4M-pick run takes about 1.8 hours, stopping at step 34,000. Correctness on this backend is not assumed: a CPU-vs-accelerator parity check on losses and gradient norms gates every launch, LayerNorm is composed from primitive ops (the fused kernel's backward is wrong on some chip/OS combinations), and a watchdog skips non-finite gradient steps and halts on recurrence. Inference exports to ONNX (opset 18) and runs on a consumer x86-64 machine, CPU-only, well under a millisecond per pick.

5 Pre-registered evaluation protocol

The evaluation is frozen at a public git tag (`eval-protocol-v1`) that predates the existence of the test data. The tag fixes the protocol document, the metric implementations, and the battery of models to be evaluated; the evaluation script refuses to run if the metric module drifts from the tagged version, and refuses any model artifact whose hash was not committed to the experiment ledger before the test data was downloaded. There is one evaluation round. Any second round would require a new public tag and disclosure of the total number of rounds.

5.1 Test set: MSH, exactly once

MSH is the first *Magic: The Gathering* set drafted on Arena after the protocol freeze: a licensed-IP (*Universes Beyond*) set, which makes the test deliberately harder (§7). The test snapshot is the first public 17Lands MSH Premier file successfully downloaded on or after its publication, frozen immediately by sha256 and ETag and never re-downloaded. Until the frozen evaluation has run, no training, tuning, metric-peeking, or feature-vocabulary fitting may read MSH pick data. MSH *card* features alone are legal (the zero-shot contract, §3). Quality gates at snapshot time: modern schema, Premier rows present, at least 2,500 expert-slice drafts, $\geq$ 99% of pack card names joining to card features. On a volume failure the single pre-declared contingency is to wait exactly one snapshot cycle. The evaluation executes within 72 hours of the gates passing: one inference pass per battery member over the frozen snapshot, cached as per-pick prediction files from which every statistic is computed. Models are never re-run during analysis.

Two populations are scored. The *expert slice* (§3) is the headline; the all-users slice is secondary. Two conditioning configurations are pre-registered: *deployment mode* (headline), conditioning on a fixed top skill bucket selected on the dev sets and frozen into the battery configuration; and *human-model mode*, conditioning on each drafter's true bucket, scored on all users. Forced picks (final pick of a pack) are included in headline numbers, matching the per-set ceilings and prior work; non-forced variants are secondary rows.

5.2 Development discipline

Every architecture, hyperparameter, text-encoder, and conditioning decision keys on one number: the unweighted mean over {BRO, TMT, SOS} of expert-slice deployment-mode top-1, computed zero-shot from a single development run (*F-dev*) trained on the universe minus those three sets and MSH. The trio is chosen for external validity: BRO is the same holdout as the only published zero-shot number [3], SOS carries our most mature per-set ceiling, and TMT is a licensed-IP set: the rehearsal for MSH's distribution shift. Early stopping never sees dev metrics (§4.3), so the scaling curve is not selected on its own y-axis.

5.3 Metrics and statistics

Primary: top-1 agreement with the human pick, expert slice, deployment mode. Secondary: top-3; all-users top-1 (human-model mode); the *normalized score*—zero-shot top-1 divided by the per-set ceiling's top-1 on identical picks (aligned pick-by-pick on the ceiling's validation split); mean log-loss; top-label expected calibration error (15 equal-mass bins, no post-hoc temperature on the test set); per-(pack, pick) accuracy curves against the per-cell random floor; and *late-draft retention*, the model/ceiling ratio restricted to picks 8+ of each pack, where pool context dominates.

All intervals are cluster bootstraps over drafts, not picks (B=2,000, fixed seed, percentile 95% CIs); paired comparisons share resample indices. Clustering matters little but is grounded empirically: measured intraclass correlation of pick correctness within drafts is $\rho = 0.0099$ (per-set SOS validation; 1,662 drafts, 69,803 picks), a design effect of ∼1.4 at ∼42 picks per draft, giving CI half-widths $\sim \pm 0.3$pp at the 2,500-expert-draft gate. Seed variance is measured with three seeds at two scaling rungs, and set-composition variance with two probe rungs (§6).

5.4 The battery

Frozen by artifact hash before the test snapshot: (1) **F-full**, the final recipe trained on all 31 sets, the headline model; its dev-set numbers are meaningless (it trains on them) and are never quoted. A pre-registered flexibility clause lets two optional extras (the Powered Cube file and VOW QuickDraft) join F-full's training data if and only if a dev-trio ablation shows a gain; the decision is frozen with the battery and reported either way. (2) **F-dev**, the development model (§above), which doubles as the top scaling rung. (3) **Scaling rungs** at 1, 2, 4, 8, 16, and 28 training sets (nested, in release order) plus two composition probes, all with the fixed recipe and a shared step cap. (4) **Ablations**: A-notext (structured features only, no text embedding) and A-noUB (the three licensed-IP training sets removed), for the shift analysis of §7. (5) **Baselines**: random; an hour-zero rarity-color heuristic; and three asterisked post-release baselines (community site ratings at day ∼2, and two 17Lands-statistics pickers) that bound what cheap post-release information buys. (6) **Post-day-one rows**, frozen now and executed only after the zero-shot evaluation: a per-set ceiling trained on the frozen MSH snapshot with the stock per-set recipe, and F-full fine-tuned on the MSH training split with hyperparameters pinned from dev-trio trial runs.

5.5 Claims bands

To keep interpretation out of our own hands, three mutually exclusive claims paragraphs (for headline results below 45%, between 45 and 55%, and above 55%) were drafted before any test number existed and are carried in the paper source; the matching one is selected verbatim when the frozen evaluation reports (§6). In all bands the secondary claims are the same: the normalized-vs-ceiling score, the shape of the scaling curve, the licensed-IP shift analysis, and the skill-conditioning effect.

6 Results

Zero-shot transfer on the development trio. Table 1 is the paper's central result: the F-dev 54.3% headline from §1, trained on 149.4M picks from 28 sets and evaluated on three sets it has never seen a pick from. Broken out per set, DraftFM reaches 48.9% / 57.5% / 56.5% on BRO / TMT / SOS against per-set ceilings of 65.6% / 71.0% / 70.2%, recovering 74.5% / 81.0% / 80.5% of those ceilings (unaligned ratio; the pre-registered normalized score on identical picks is [PENDING: aligned normalized scores, dev trio]). Excluding forced picks lowers the three numbers to 45.2% / 54.1% / 53.2%. The deployment-mode skill bucket selected on dev is the 0.66 win-rate bucket.

Reading the BRO row. BRO is the holdout on which Bertram et al. [3] report 55.44%: the strongest published unseen-set number, achieved with card features, learned image embeddings, *and* 16 usage statistics computed from human play after the set's release. Our 48.9% is below it. The comparison the BRO row is designed for is the day-one setting, where usage statistics do not exist: against the same paper's features-only representation (33.6–35.6%, single-set training) and every day-one-feasible baseline in Table 2, multi-set training on public card features closes most of the distance toward the usage-statistics-inclusive number while using strictly release-day information. BRO is also the trio's hardest set for transfer: its packs mix in a 63-card retro-artifact bonus sheet. The gap between the BRO and TMT/SOS rows – and why TMT, the trio's licensed-IP dev set, nonetheless transfers as well as SOS – is analysed in §7; Figure 4 shows the same interface on TMT. For sets in the training corpus but without a dedicated per-set model, the deployed system instead falls back to F-full's scoring; Figure 5 shows what that looks like today for LTR, LCI, and DMU – a deployment illustration, not an additional zero-shot data point.

Baselines. Table 2: random picking scores 23.2%, and an hour-zero heuristic (prefer lower rarity for playables, follow the pool's colors) reaches 42.8%: just below GPT-4o's 43% [2] and close to the expert-tuned 44.5% of Ward et al. [10], a useful caution about how much of drafting is surface regularity. DraftFM's zero-shot numbers, on each of the three dev sets, clear every day-one-feasible baseline in the table (Random and the rarity heuristic are measured on SOS as a rehearsal stand-in, not per-set). The three asterisked post-release baselines bound what cheap human data buys within days of release. Their MSH rows are filled in once the frozen evaluation runs.

The frozen MSH evaluation. The headline: F-full expert-slice deployment-mode top-1 on MSH is [PENDING: F-full MSH top-1]% (CI in Table 1). Snapshot gates, identifiers, and hashes are reported in Appendix B. [PENDING: MSH secondary metrics: top-3, all-users top-1, normalized score vs the post-day-one MSH ceiling, log-loss, ECE, late-draft retention]

Scaling. Figure 6 and Table 3 report zero-shot accuracy as training grows from one set (NEO, 5.4M picks) to the full 28-set universe (149.4M picks), under the fixed recipe. Single-set training is the regime of Bertram et al. [3]'s features-only anchor; the top rung is F-dev. Two extra seeds at S1 and S4 give within-training top-1 spreads of 0.677–0.682 and 0.688–0.689 – training itself is not noise-dominated at either rung, though this checks in-distribution stability, not a seed sweep of the zero-shot dev-mean the curve actually plots. [PENDING: scaling-curve shape claim: slope,

Table 1: Zero-shot expert-slice top-1 agreement (%), deployment mode, forced picks included; 95% cluster-bootstrap CIs over drafts in parentheses. F-full's dev cells are never quoted (it trains on those sets). Each cell's source run is recorded as a comment in the generated table file.

Model	BRO	TMT	SOS	Dev mean	MSH
DraftFM (F-dev, zero-shot)	48.9 (48.8, 49.0)	57.5 (57.3, 57.7)	56.5 (56.4, 56.6)	54.3	TBD
DraftFM (F-full, zero-shot)	–	–	–	–	TBD
Per-set ceiling	65.6	71.0	70.2 (69.8, 70.6)	68.9	TBD
F-dev / ceiling (unaligned)	0.745	0.810	0.805	0.787	TBD

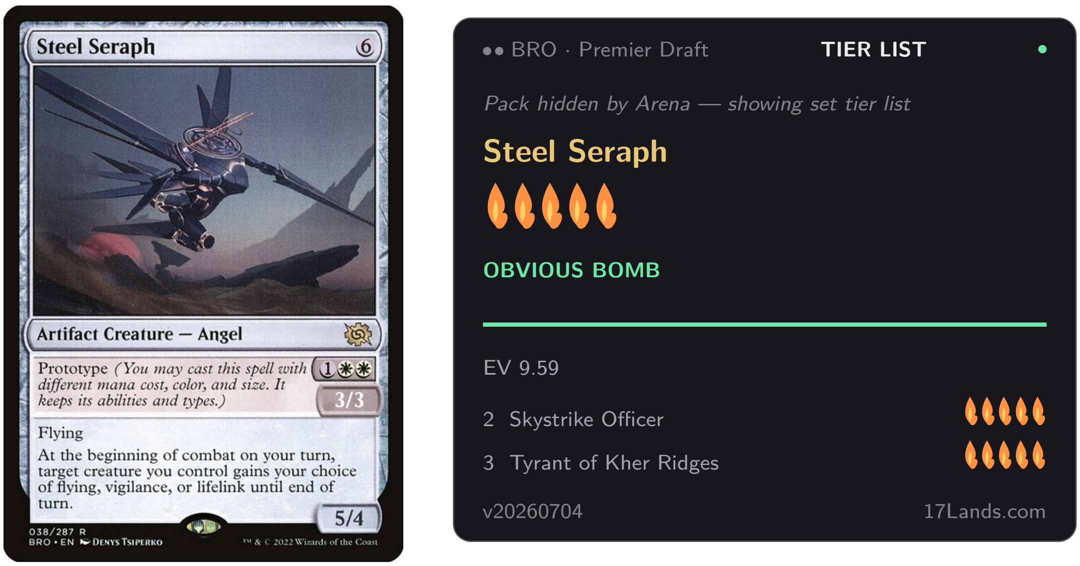

Figure 3: The same interface in *tier-list* mode (no live pack; Arena hides the pack between picks), on BRO, the dev trio's hardest transfer set. Flames come from the card's percentile rank in the whole set's P1P1 expected value, not a head-to-head split, so no percentage is shown (§4.2). Steel Seraph's real EV (9.59) sits at the 99.4th percentile of BRO's card pool; the two runners-up sit at the 99.0th and 98.7th. Computed by the deployed per-set model, never MSH.

diminishing-returns point, and whether set count or pick count is the better predictor (S2b/S4b probes + 3-seed bands at S1/S4)]

7 Analysis

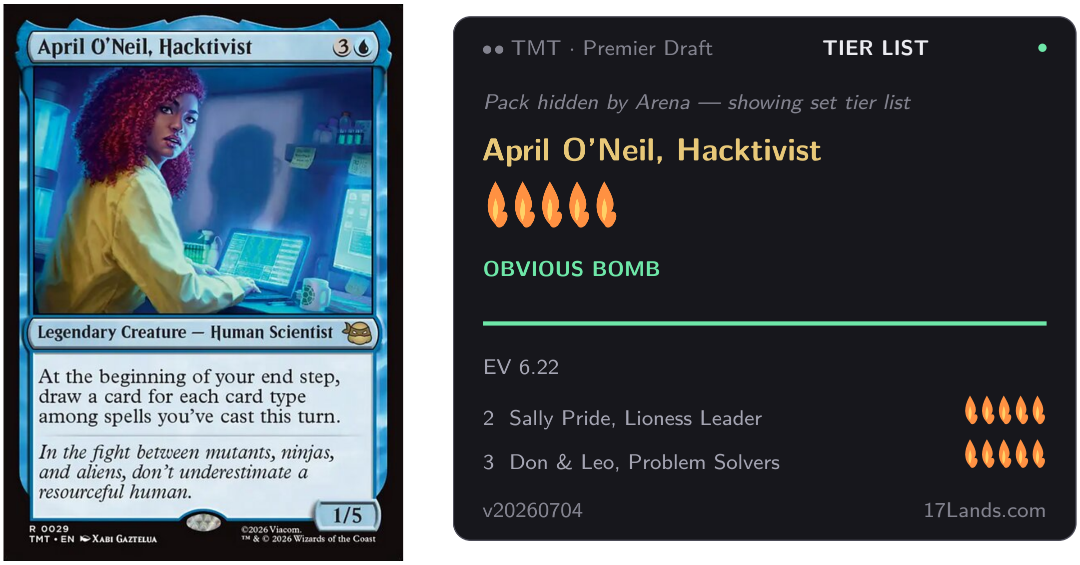

Figure 4: The same interface on TMT, the dev trio's licensed-IP set (§7) – a comic-book bomb the text encoder has never seen described in *Magic: The Gathering* rules-text style. April O'Neil's real EV (6.22) sits at the 96.7th percentile of TMT's pool. Computed by the deployed per-set model, never MSH.

Figure 5: Three sets with no dedicated per-set model – Lord of the Rings, The Lost Caverns of Ixalan, Dominaria United – all in F-full's 31-set training corpus. This is what the deployed system actually serves for them today: F-full as the fallback whenever a set has no dedicated per-set model, the same fallback path it would run for a set onboarded on day one. Real EVs and percentiles, computed the same way as Figures 3 and 4 – but because F-full trained on these three sets, this illustrates the production fallback path, not a zero-shot-accuracy claim; that claim is Table 1's dev-trio and MSH rows only.

Table 2: Baselines and published anchors. Upper block: our measured baselines (practice rows will be replaced by frozen MSH rows by the same pipeline). Lower block: published numbers, quoted from their papers; test sets and information regimes differ, so these situate rather than compare. * = uses post-release or otherwise non-day-one information.

Method	Information used	Test set	Expert top-1 (%)
Random	none	SOS PremierDraft (rehearsal)	23.2 (23.2, 23.3)
Rarity–color heuristic	card features (hour 0)	SOS PremierDraft (rehearsal)	42.8 (42.7, 42.9)
Site-ratings heuristic (day ~2)*	post-release statistics	MSH	TBD
ALSA-argmin*	post-release statistics	MSH	TBD
shrunk-GIH-argmax*	post-release statistics	MSH	TBD
DraftsimBot [10]	expert-tuned ratings	M19 (Draftsim)	44.5
GPT-4o zero-shot* [2]	LLM pretraining	NEO	43.0
Features-only, single-set [3]	card features	unseen (NEO-trained)	33.6–35.6
Features+meta+image, 13-set* [3]	post-release meta + images	BRO (unseen)	55.4

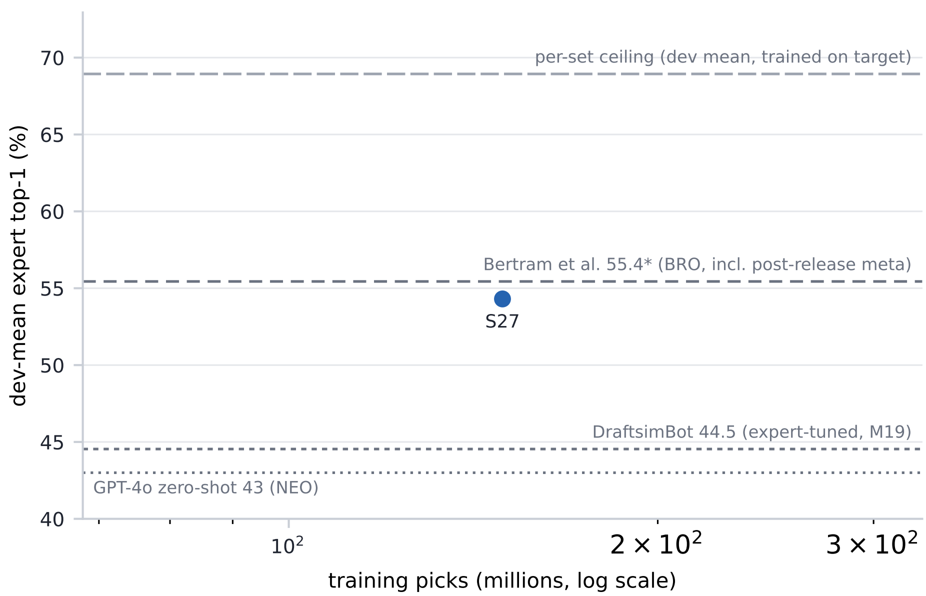

Figure 6: Zero-shot dev-trio-mean expert top-1 vs training picks (log scale). Filled points: the nested scaling ladder; open points: set-composition probes. Horizontal rules: published anchors and the per-set ceiling mean. Generated from run records by `figures/scaling_curve.py`; rungs appear as their frozen evaluations land.

Table 3: Scaling ladder. Rung labels are frozen protocol identifiers; set and pick counts come from run records. Dev-mean is the model-selection metric of §5.

Rung	Sets	#sets	Train picks (M)	Best step	Dev-mean top-1 (%)
S1	NEO	1	5.4	4000	48.9
S2	S1 + DSK	2	13.3	18000	50.9
S2b	MOM, TDM (probe)	2	14.5	8000	50.2
S4	S2 + DMU, FIN	4	27.8	24000	51.7
S4b	STX, SNC, OTJ, TLA (probe)	4	23.5	20000	51.0
S8	S4 + STX, MOM, BLB, TLA	8	52.9	18000	52.4
S16	S8 + AFR, SNC, ONE, LTR, WOE, MKM, OTJ, EOE	16	99.9	34000	53.3
S27	full F-dev universe	28	149.4	34000	54.3

Seed variance on the top-scoring ablation. Two additional seeds of A-noctx (the recipe F-full uses, §4.2) land within-training top-1 at 0.680–0.684, a 0.5pp spread around the seed reported in the table – A-noctx's own accuracy is not a single lucky initialization, though the close margins between it and the neighboring recipe-search variants (54.2–54.8 dev-mean, Table 4) mean the full ranking has not been seed-checked.

The licensed-IP shift. MSH's names and flavor come from a comic universe, not from *Magic: The Gathering*. Its text lives in the sentence encoder's pretraining as a different genre. This is the distribution shift most likely to break a text-dependent model on the day it matters, and it is why the protocol makes a licensed-IP set (TMT) a development set: the rehearsal happens where it can still inform design. One mitigation is built in: card names are masked out of the text before embedding (§4.1), so the encoder sees mechanics, not franchises. A second design choice does not reduce the shift but makes it measurable: the ablation grid of Table 4 separates what the text embedding contributes on in-universe sets (SOS, BRO) from what it contributes on licensed sets (TMT, MSH). Zero-shot, the licensed dev set is not the hard one: TMT reaches 57.5% against SOS's 56.5% (and 81.0% vs 80.5% of the respective ceilings), suggesting the masking plus structured features carry the load. Whether that holds because text adds little on licensed sets or because TMT is simply an easier format is what the grid decides: [PENDING: A-notext/A-noUB deltas per set and the UB-penalty estimate]. If the test set is harder than the trio average, the pre-registered framing stands: a licensed-IP test makes the zero-shot claim more credible, not less.

Why BRO transfers worst. BRO sits [PENDING: BRO gap vs trio mean] below the other dev sets and only 74.5% of its ceiling. Three candidate explanations are checkable in the per-pick curves: its packs mix in a 63-card retro-artifact bonus sheet drawn from a much older design idiom, inflating the effective card pool; artifact-heavy formats weaken color-signal features that transfer well elsewhere; and its 2022 play patterns are the furthest in time from the corpus median. [PENDING: per-(pack,pick) curve comparison and bonus-sheet-slice accuracy]

Skill conditioning. Conditioning is a scorer-side input (§4.2), so one model serves any audience. In human-model mode (§5) the model scores 48.4% / 57.8% / 56.8% on BRO / TMT / SOS: within a point of the expert-slice deployment numbers, consistent with skill moving picks less than card quality does. The pre-registered effect size (deployment-mode accuracy as a function of the conditioning bucket, and the gap between conditioning on the top bucket versus the drafter's own) is [PENDING: skill-conditioning effect on MSH].

Calibration. Zero-shot confidence falls well short of within-set calibration: top-label ECE is 0.102 / 0.071 / 0.093 on the dev trio, against 0.014 for the within-set SOS ceiling, roughly a five- to seven-and-a-half-fold gap. The protocol permits one calibration decision: a temperature fitted on dev only and frozen into the battery, never fitted on MSH. The frozen choice and MSH ECE are [PENDING: temperature decision + MSH ECE].

Late-draft retention. Later picks are where a set-agnostic model could plausibly collapse to rate-the-card, since pool signal has taken over from the card alone. The pre-registered statistic, late-draft retention (§5), is [PENDING: late-draft retention, dev trio and MSH], alongside the per-(pack, pick) curves against the per-cell random floor.

Table 4: Zero-shot expert top-1 (%), point estimates (per-cell 95% CI half-widths are $\sim \pm$ 0.1–0.2pp on the dev sets, Table 1; paired-difference CIs accompany the deltas in the text). A-noctx, A-proportional, and A-topfilter are recipe-search variants against the Full baseline, run before the deployed architecture was fixed; A-notext and A-noUB are the pre-registered ablations (§5) analysed below. A-notext drops the 384-d text embedding from the Full recipe. A-noUB removes the three licensed-IP training sets from A-noctx (the winning architecture, §4.2), so its clean comparator is the A-noctx row, not Full. The (variant × set) grid supplies the dev-side numbers for the licensed-IP text penalty (per-set deltas below), computed before MSH is seen.

Variant	BRO	TMT	SOS	Dev mean	MSH
Full (F-dev recipe)	48.9	57.5	56.5	54.3	TBD
A-notext	45.8	55.9	53.1	51.6	TBD
A-noctx (winning)	50.0	57.7	56.6	54.8	TBD
A-proportional	50.2	56.8	55.7	54.2	TBD
A-topfilter	49.8	57.3	56.5	54.5	TBD
A-noUB	49.2	57.5	54.9	53.9	TBD

Acknowledgments

This work would not exist without 17Lands, whose public datasets [1] are the training and evaluation corpus, and whose data-sharing policy makes reproducible drafting research possible. Card metadata is provided by Scryfall [9]. *Magic: The Gathering* is the property of Wizards of the Coast; this is unofficial research, unaffiliated with and unendorsed by Wizards of the Coast, 17Lands, or Scryfall.

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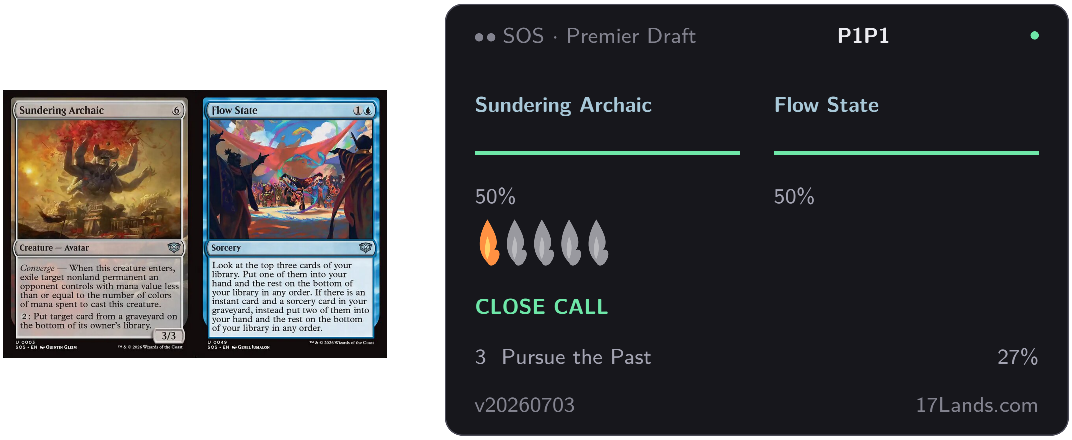

Figure 7: The low-conviction end of the same interface: a real thirteen-card SOS pack (empty pool) where the model's top two EVs are (1.41 vs. 1.41) a near-exact tie, so the panel switches to a head-to-head duo view (§4.2) instead of naming a single pick. Dominance this close (50/50) is the regime top-label ECE below is measuring: the model's stated confidence and its actual reliability at low conviction. Computed by the deployed per-set model, never MSH.

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A Limitations

The target is agreement, not optimality. Top-1 agreement with expert humans measures imitation. Experts disagree with each other (the within-set ceiling is ∼70%, not 100%), and agreeing with them is not the same as maximising win rate. A model could in principle draft better than its agreement suggests, or worse. Win-rate evaluation of zero-shot drafters is future work.

Population. 17Lands users are self-selected: they run a tracker, they draft on Arena, and the expert slice further conditions on volume and success. All numbers are relative to this population; paper drafting, other platforms, and casual play are out of scope.

One evaluation round. Pre-registration buys credibility at the cost of adaptivity: the frozen battery cannot react to surprises in the test snapshot, and a single set is a sample of one from the distribution of future sets. The dev-trio spread and the composition probes bound, but do not eliminate, set-to-set variance in what the headline number would be on the *next* set.

Text and language. The text encoder is English-only and frozen. Rules text is embedded as prose, so mechanics whose meaning is mostly numeric or positional may be under-represented. Name masking removes the most direct pretraining leak for licensed sets, but flavor vocabulary in the remaining text still differs by universe (§7).

Numerics. Training runs on an accelerator backend with documented silent-error classes; we gate on CPU parity and skip non-finite steps (§4.3), which guards the classes we know about.

B Reproducibility

Every result in this paper traces to a run id in an append-only ledger, and every table cell and prose number is emitted by a script from that ledger and cached evaluation outputs – no result number here is typed by hand. The ledger schema, frozen identifiers (protocol tag, manifest hashes), release contents, and script-level pointers for reproducing any specific number are in the companion implementation reference, not repeated here.

C Licensing and attribution

Draft data. Draft data is from the 17Lands public datasets [1], used under CC BY 4.0 and per the project's usage guidelines (bulk S3 downloads; no API scraping).

Card metadata. Card metadata is from Scryfall [9], used per its API guidelines; Scryfall neither produces nor endorses this work.

Card images. Figures 1, 3, 7, 4, and 5 reproduce official Scryfall card scans [9], fetched directly from Scryfall's own hosted image URLs (fetch script and source manifest released with the companion reference). Scryfall's stated policy on card images ("Use of Scryfall Data and Images," `https://scryfall.com/docs/api`) permits this use, subject to rules we follow throughout: every image is shown full and unmodified, with no cropping, distorting, blurring, sharpening, desaturating, color-shifting, or watermarking of the card art itself. Any score or label in a figure sits beside the image, never on top of it. Because the full card image is always used (never an art crop), the copyright and artist line Scryfall prints at the bottom of the card remains visible in every figure, and nothing here implies that Wizards of the Coast produced or endorses this work. The cards illustrated are drawn from BRO, SOS, TMT, LTR, LCI, and DMU – all public, released sets (§3) – never from MSH, which remains untouched until the frozen evaluation of §5.

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